

LAB No. 17

Health Monitoring of Disk in Linux Using Shell Scripting Automation

In modern IT environments, disk failures and storage issues can lead to data loss, system downtime, and performance degradation. Continuous monitoring of disk health is essential for system reliability.

This project focuses on automating disk health monitoring in Linux using shell scripting. The system periodically checks disk usage, disk errors, and SMART health status of storage devices. When predefined thresholds are exceeded (for example, disk usage above 80% or disk health warnings), the script generates alerts through log files or email notifications.

The automation is achieved using shell scripts combined with cron jobs, enabling proactive disk management without manual intervention.

LAB Objectives

- Monitor disk usage and available storage space
- Automate periodic health checks using cron
- Generate alerts for abnormal disk conditions
- Maintain log files for future analysis

Tools and Software Required

Operating System

- Linux (Ubuntu / CentOS / Red Hat / Debian)

Disk Usage Monitoring

The df command reports:

- Total disk space
- Used space
- Available space
- Usage percentage

Monitoring disk usage helps prevent:

- Application crashes

- System freezes
- Log file overflow

Shell Scripting Automation

Shell scripting allows:

- Execution of system commands automatically
- Decision making using conditions (if, case)
- Looping over multiple disks
- Logging output to files

Automation reduces human error and ensures consistency.

LAB Task:

To make a shell script to perform automation for system health monitoring

(Note: Students need to attach your screenshots and working here)

SOL:

```
GNU nano 7.2 health_monitor.sh *
#!/bin/bash

# Set threshold
THRESHOLD=80

# Log file
LOGFILE="$HOME/health_log.txt"

echo "==== System Health Check =====> $LOGFILE
date >> $LOGFILE

echo "Disk Usage:" >> $LOGFILE

# Check disk usage
df -h | while read line
do
    usage=$(echo $line | awk '{print $5}' | sed 's/%//g')

    if [ "$usage" -gt "$THRESHOLD" ] 2>/dev/null
    then
        echo "WARNING: High Disk Usage -> $line" >> $LOGFILE
    fi
done

# Memory usage
echo "Memory Usage:" >> $LOGFILE
free -h >> $LOGFILE
```

```
# CPU Load
echo "CPU Load:" >> $LOGFILE
uptime >> $LOGFILE

echo "===== " >> $LOGFILE
echo "" >> $LOGFILE
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ nano health_monitor.sh
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ chmod +x health_monitor.sh
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ./health_monitor.sh
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cat ~/health_log.txt
===== System Health Check =====
Sun May 3 03:00:30 PM PKT 2026
Disk Usage:
WARNING: High Disk Usage -> /dev/sr1      6.0G  6.0G   0 100% /media/abdullah-sohail/Ubuntu 24.04.3 LTS amd64
WARNING: High Disk Usage -> /dev/sr0      96M   96M   0 100% /media/abdullah-sohail/CDROM
Memory Usage:


|       | total | used  | free  | shared | buff/cache | available |
|-------|-------|-------|-------|--------|------------|-----------|
| Mem:  | 2.8Gi | 1.3Gi | 94Mi  | 43Mi   | 1.6Gi      | 1.5Gi     |
| Swap: | 2.8Gi | 316Ki | 2.8Gi |        |            |           |


CPU Load:
 15:00:30 up 3 min,  1 user,  load average: 1.05, 1.00, 0.47
=====
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ crontab -e
crontab: installing new crontab
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
*/1 * * * * /bin/bash /home/abdullah-sohail/health_monitor.sh
```

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cat ~/health_log.txt
===== System Health Check =====
Sun May  3 03:00:30 PM PKT 2026
Disk Usage:
WARNING: High Disk Usage -> /dev/sr1          6.0G  6.0G    0 100% /media/abdullah-sohail/Ubuntu 24.04.3 LTS amd64
WARNING: High Disk Usage -> /dev/sr0          96M   96M    0 100% /media/abdullah-sohail/CDROM
Memory Usage:
      total      used      free    shared  buff/cache   available
Mem:    2.8Gi    1.3Gi    94Mi     43Mi     1.6Gi     1.5Gi
Swap:    2.8Gi    316Ki    2.8Gi
CPU Load:
 15:00:30 up 3 min,  1 user,  load average: 1.05, 1.00, 0.47
=====

```

```

===== System Health Check =====
Sun May  3 03:04:01 PM PKT 2026
Disk Usage:
WARNING: High Disk Usage -> /dev/sr1          6.0G  6.0G    0 100% /media/abdullah-sohail/Ubuntu 24.04.3 LTS amd64
WARNING: High Disk Usage -> /dev/sr0          96M   96M    0 100% /media/abdullah-sohail/CDROM
Memory Usage:
      total      used      free    shared  buff/cache   available
Mem:    2.8Gi    1.2Gi   180Mi     40Mi     1.6Gi     1.6Gi
Swap:    2.8Gi    316Ki    2.8Gi
CPU Load:
 15:04:01 up 7 min,  1 user,  load average: 0.05, 0.54, 0.39
=====

```

```

===== System Health Check =====
Sun May  3 03:05:01 PM PKT 2026
Disk Usage:
WARNING: High Disk Usage -> /dev/sr1          6.0G  6.0G    0 100% /media/abdullah-sohail/Ubuntu 24.04.3 LTS amd64
WARNING: High Disk Usage -> /dev/sr0          96M   96M    0 100% /media/abdullah-sohail/CDROM

```

```

===== System Health Check =====
Sun May  3 03:06:01 PM PKT 2026
Disk Usage:
WARNING: High Disk Usage -> /dev/sr1          6.0G  6.0G    0 100% /media/abdullah-sohail/Ubuntu 24.04.3 LTS amd64
WARNING: High Disk Usage -> /dev/sr0          96M   96M    0 100% /media/abdullah-sohail/CDROM
Memory Usage:
      total      used      free    shared  buff/cache   available
Mem:    2.8Gi    1.2Gi   179Mi     40Mi     1.6Gi     1.6Gi
Swap:    2.8Gi    316Ki    2.8Gi
CPU Load:
 15:06:01 up 9 min,  1 user,  load average: 0.19, 0.45, 0.38
=====

```

```

===== System Health Check =====
Sun May  3 03:07:01 PM PKT 2026
Disk Usage:
WARNING: High Disk Usage -> /dev/sr1          6.0G  6.0G    0 100% /media/abdullah-sohail/Ubuntu 24.04.3 LTS amd64
WARNING: High Disk Usage -> /dev/sr0          96M   96M    0 100% /media/abdullah-sohail/CDROM
Memory Usage:
      total      used      free    shared  buff/cache   available
Mem:    2.8Gi    1.2Gi   181Mi     40Mi     1.6Gi     1.6Gi
Swap:    2.8Gi    316Ki    2.8Gi
CPU Load:
 15:07:01 up 10 min, 1 user,  load average: 0.07, 0.37, 0.35
=====

```

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ █

```